

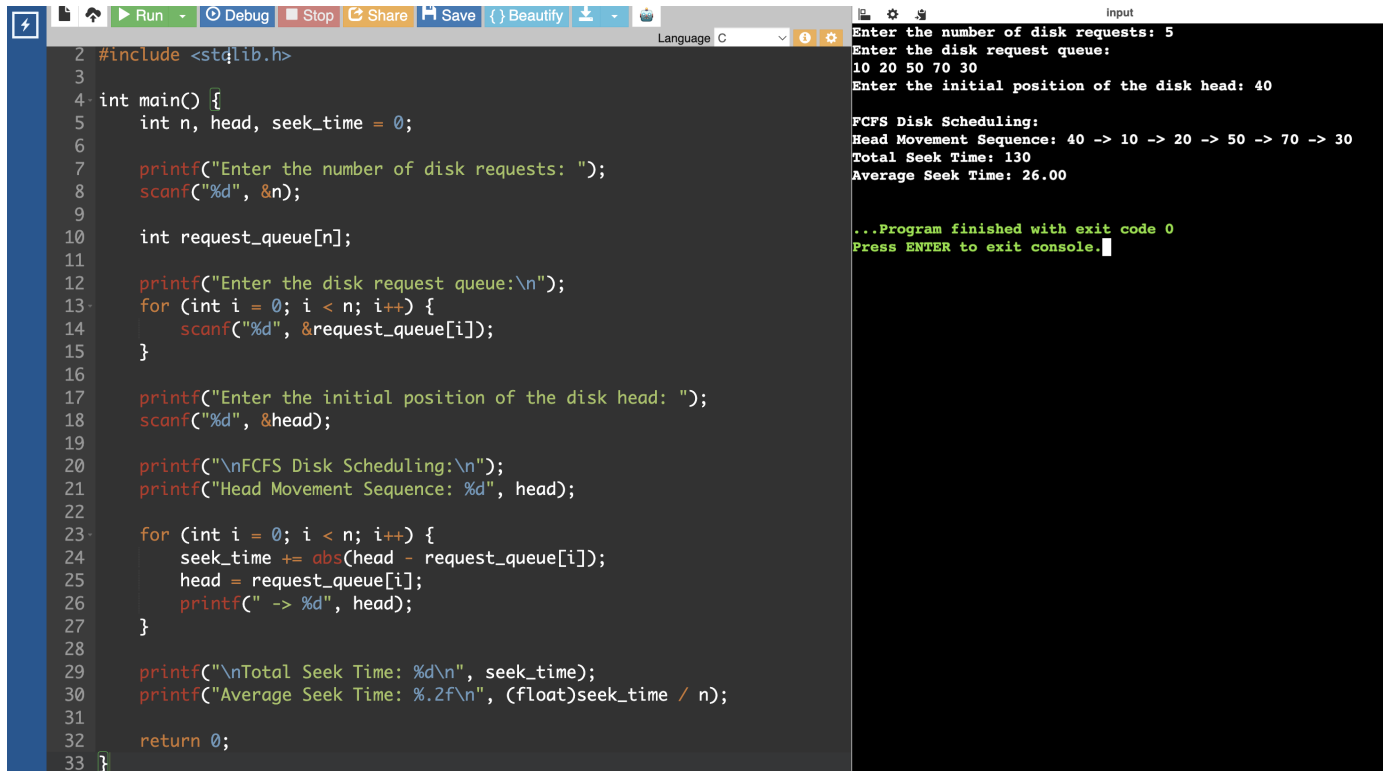
CSA0404 – OPERATING SYSTEMS

EXPERIMENT 37

QUESTION

Construct a C program to simulate the First Come First Served disk scheduling algorithm.

PROGRAM



```
2 #include <stdlib.h>
3
4 int main() {
5     int n, head, seek_time = 0;
6
7     printf("Enter the number of disk requests: ");
8     scanf("%d", &n);
9
10    int request_queue[n];
11
12    printf("Enter the disk request queue:\n");
13    for (int i = 0; i < n; i++) {
14        scanf("%d", &request_queue[i]);
15    }
16
17    printf("Enter the initial position of the disk head: ");
18    scanf("%d", &head);
19
20    printf("\nFCFS Disk Scheduling:\n");
21    printf("Head Movement Sequence: %d", head);
22
23    for (int i = 0; i < n; i++) {
24        seek_time += abs(head - request_queue[i]);
25        head = request_queue[i];
26        printf(" -> %d", head);
27    }
28
29    printf("\nTotal Seek Time: %d\n", seek_time);
30    printf("Average Seek Time: %.2f\n", (float)seek_time / n);
31
32    return 0;
33 }
```

Input

```
Enter the number of disk requests: 5
Enter the disk request queue:
10 20 50 70 30
Enter the initial position of the disk head: 40

FCFS Disk Scheduling:
Head Movement Sequence: 40 -> 10 -> 20 -> 50 -> 70 -> 30
Total Seek Time: 130
Average Seek Time: 26.00

...Program finished with exit code 0
Press ENTER to exit console.
```